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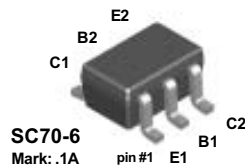
# FFB3904 / FMB3904 / MMPQ3904

## NPN Multi-Chip General Purpose Amplifier

### Description

This device is designed as a general-purpose amplifier and switch. The useful dynamic range extends to 100 mA as a switch and to 100 MHz as an amplifier. Sourced from Process 23.

### Block Diagram



The pinouts are symmetrical; pin 1 and pin 4 are interchangeable. Units inside the carrier tape can be of either orientation (0 deg and 180 deg) and will not affect the functionality of the device.

Figure 1. FFB3904 Device Package

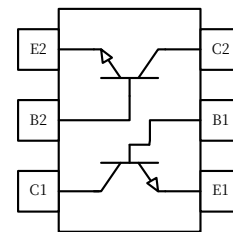


Figure 2. FFB3904 Internal Connection

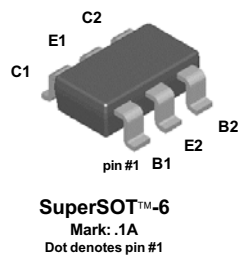


Figure 3. FMB3904 Device Package

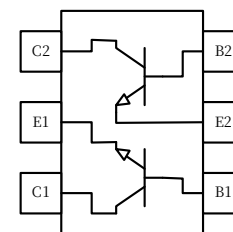


Figure 4. FMB3904 Internal Connection

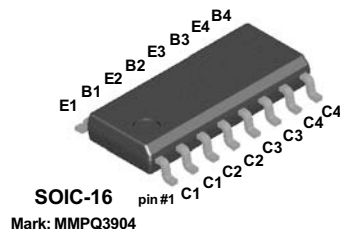


Figure 5. MMPQ3904 Device Package

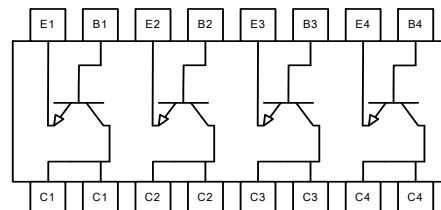


Figure 6. MMPQ3904 Internal Connection

## Ordering Information

| Part Number | Top Mark | Package  | Packing Method |
|-------------|----------|----------|----------------|
| FFB3904     | .1A      | SC70 6L  | Tape and Reel  |
| FMB3904     | .1A      | SSOT 6L  | Tape and Reel  |
| MMPQ3904    | MMPQ3904 | SOIC 16L | Tape and Reel  |

## Absolute Maximum Ratings<sup>(1)</sup>

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only. Values are at  $T_A = 25^\circ\text{C}$  unless otherwise noted.

| Symbol         | Parameter                                        | Value       | Unit             |
|----------------|--------------------------------------------------|-------------|------------------|
| $V_{CEO}$      | Collector-Emitter Voltage                        | 40          | V                |
| $V_{CBO}$      | Collector-Base Voltage                           | 60          | V                |
| $V_{EBO}$      | Emitter-Base Voltage                             | 6.0         | V                |
| $I_C$          | Collector Current - Continuous                   | 200         | mA               |
| $T_J, T_{STG}$ | Operating and Storage Junction Temperature Range | -55 to +150 | $^\circ\text{C}$ |

### Note:

1. These ratings are based on a maximum junction temperature of  $150^\circ\text{C}$ . These are steady-state limits. ON Semiconductor should be consulted on applications involving pulsed or low-duty cycle operations.

## Thermal Characteristics<sup>(2)</sup>

Values are at  $T_A = 25^\circ\text{C}$  unless otherwise noted.

| Symbol          | Parameter                                                | Max.    |         |          | Unit                      |
|-----------------|----------------------------------------------------------|---------|---------|----------|---------------------------|
|                 |                                                          | FFB3904 | FMB3904 | MMPQ3904 |                           |
| $P_D$           | Total Device Dissipation                                 | 300     | 700     | 1,000    | mW                        |
|                 | Derate above $25^\circ\text{C}$                          | 2.4     | 5.6     | 8.0      | mW/ $^\circ\text{C}$      |
| $R_{\theta JA}$ | Thermal Resistance, Junction to Ambient                  | 415     | 180     |          | $^\circ\text{C}/\text{W}$ |
|                 | Thermal Resistance, Junction to Ambient, Effective 4 Die |         |         | 125      |                           |
|                 | Thermal Resistance, Junction to Ambient, Each Die        |         |         | 240      |                           |

### Note:

2. PCB size: FR-4 76 x 114 x 0.6T mm<sup>3</sup> (3.0 inch x 4.5 inch x 0.062 inch) with minimum land pattern size.

## Electrical Characteristics

Values are at  $T_A = 25^\circ\text{C}$  unless otherwise noted.

| Symbol                                       | Parameter                            |                                                 | Conditions                                                  | Min. | Typ. | Max. | Unit |
|----------------------------------------------|--------------------------------------|-------------------------------------------------|-------------------------------------------------------------|------|------|------|------|
| Off Characteristics                          |                                      |                                                 |                                                             |      |      |      |      |
| V <sub>(BR)CEO</sub>                         | Collector-Emitter Breakdown Voltage  |                                                 | I <sub>C</sub> = 1.0 mA, I <sub>B</sub> = 0                 | 40   |      |      | V    |
| V <sub>(BR)CBO</sub>                         | Collector-Base Breakdown Voltage     |                                                 | I <sub>C</sub> = 10 μA, I <sub>E</sub> = 0                  | 60   |      |      | V    |
| V <sub>(BR)EBO</sub>                         | Emitter-Base Breakdown Voltage       |                                                 | I <sub>E</sub> = 10 μA, I <sub>C</sub> = 0                  | 6.0  |      |      | V    |
| I <sub>BL</sub>                              | Base Cut-Off Current                 |                                                 | V <sub>CE</sub> = 30 V, V <sub>BE</sub> = -3 V              |      |      | 50   | nA   |
| I <sub>CEX</sub>                             | Collector Cut-Off Current            |                                                 | V <sub>CE</sub> = 30 V, V <sub>BE</sub> = -3 V              |      |      | 50   | nA   |
| On Characteristics <sup>(3)</sup>            |                                      |                                                 |                                                             |      |      |      |      |
| h <sub>FE</sub>                              | DC Current Gain                      | FFB3904, FMB3904                                | I <sub>C</sub> = 0.1 mA, V <sub>CE</sub> = 1.0 V            | 40   |      |      |      |
|                                              |                                      | MMPQ3904                                        |                                                             | 30   |      |      |      |
|                                              |                                      | FFB3904, FMB3904                                | I <sub>C</sub> = 1.0 mA, V <sub>CE</sub> = 1.0 V            | 70   |      |      |      |
|                                              |                                      | MMPQ3904                                        |                                                             | 50   |      |      |      |
|                                              |                                      | FFB3904, FMB3904                                | I <sub>C</sub> = 10 mA, V <sub>CE</sub> = 1.0 V             | 100  |      | 300  |      |
|                                              |                                      | MMPQ3904                                        |                                                             | 75   |      |      |      |
|                                              |                                      | All Devices                                     | I <sub>C</sub> = 50 mA, V <sub>CE</sub> = 1.0 V             | 60   |      |      |      |
|                                              |                                      | All Devices                                     | I <sub>C</sub> = 100 mA, V <sub>CE</sub> = 1.0 V            | 30   |      |      |      |
| V <sub>CE(sat)</sub>                         | Collector-Emitter Saturation Voltage | I <sub>C</sub> = 10 mA, I <sub>B</sub> = 1.0 mA |                                                             |      | 0.2  | V    |      |
|                                              |                                      | I <sub>C</sub> = 50 mA, I <sub>B</sub> = 5.0 mA |                                                             |      | 0.3  |      |      |
| V <sub>BE(sat)</sub>                         | Base-Emitter Saturation Voltage      | I <sub>C</sub> = 10 mA, I <sub>B</sub> = 1.0 mA | 0.65                                                        |      | 0.85 | V    |      |
|                                              |                                      | I <sub>C</sub> = 50 mA, I <sub>B</sub> = 5.0 mA |                                                             |      | 0.95 |      |      |
| Small-Signal Characteristics (MMPQ3904 only) |                                      |                                                 |                                                             |      |      |      |      |
| f <sub>T</sub>                               | Current Gain-Bandwidth Product       |                                                 | I <sub>C</sub> = 10 mA, V <sub>CE</sub> = 20 V, f = 100 MHz |      | 250  |      | MHz  |
| C <sub>ob</sub>                              | Output Capacitance                   |                                                 | V <sub>CB</sub> = 5.0 V, I <sub>E</sub> = 0, f = 140 kHz    |      | 4.0  |      | pF   |
| C <sub>ib</sub>                              | Input Capacitance                    |                                                 | V <sub>BE</sub> = 0.5 V, I <sub>C</sub> = 0, f = 140 kHz    |      | 8.0  |      | pF   |

### Note:

3. Pulse test: pulse width  $\leq 300\text{ }\mu\text{s}$ , duty cycle  $\leq 2.0\%$ .

## Typical Performance Characteristics

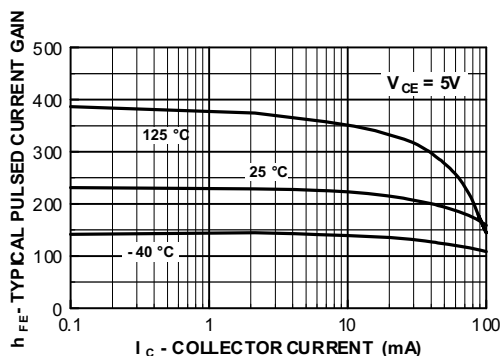


Figure 7. Typical Pulsed Current Gain vs. Collector Current

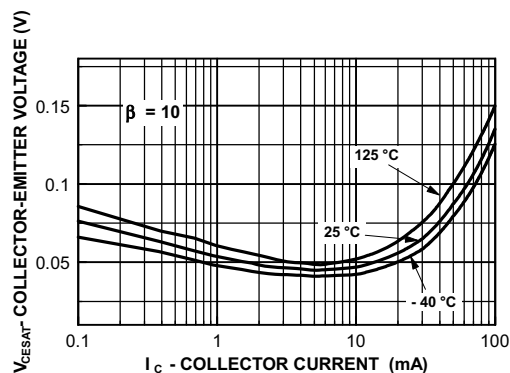


Figure 8. Collector-Emitter Saturation Voltage vs. Collector Current

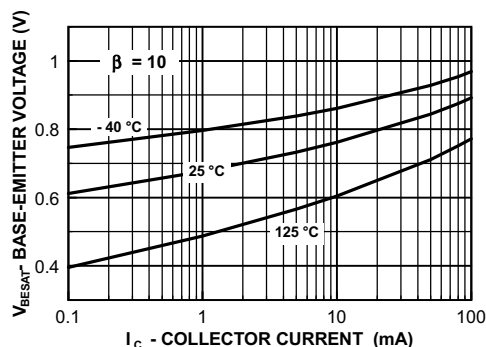


Figure 9. Base-Emitter Saturation Voltage vs. Collector Current

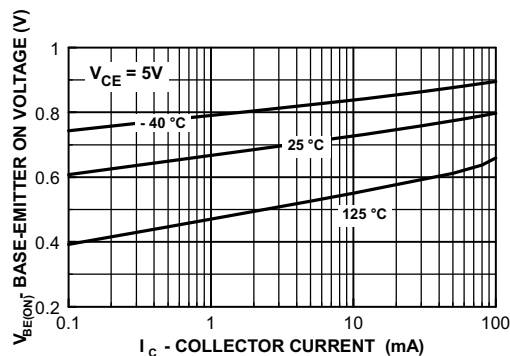


Figure 10. Base-Emitter On Voltage vs. Collector Current

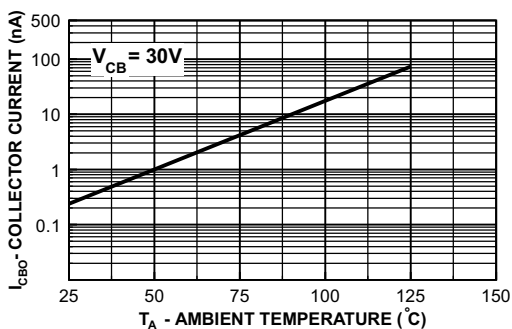


Figure 11. Collector Cut-Off Current vs. Ambient Temperature

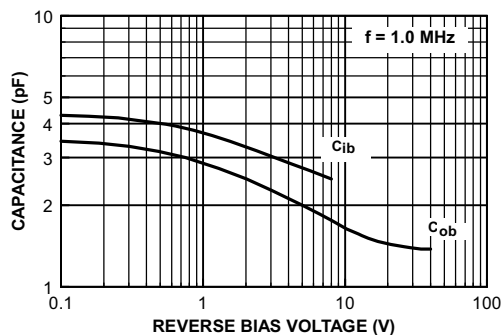


Figure 12. Capacitance vs. Reverse Bias Voltage

# Typical Performance Characteristics (Continued)

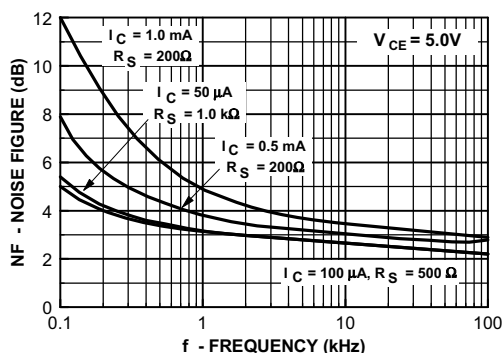


Figure 13. Noise Figure vs. Frequency

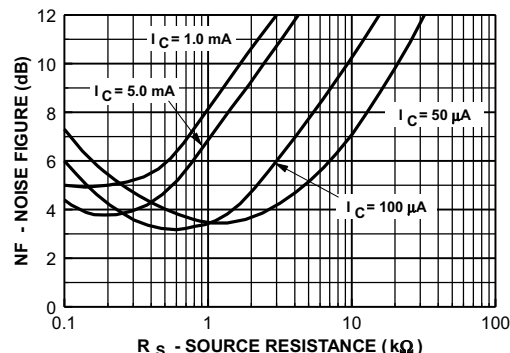


Figure 14. Noise Figure vs. Source Resistance

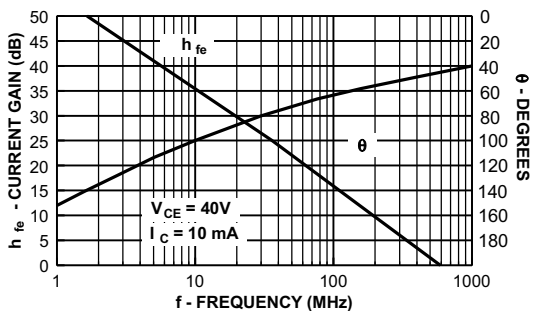


Figure 15. Current Gain and Phase Angle vs. Frequency

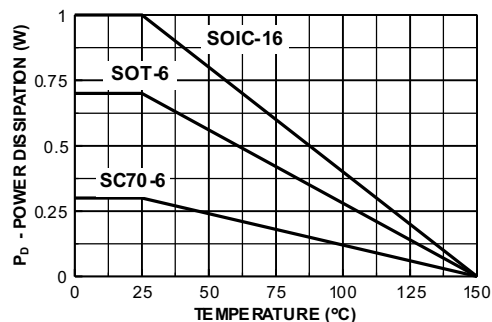


Figure 16. Power Dissipation vs. Ambient Temperature

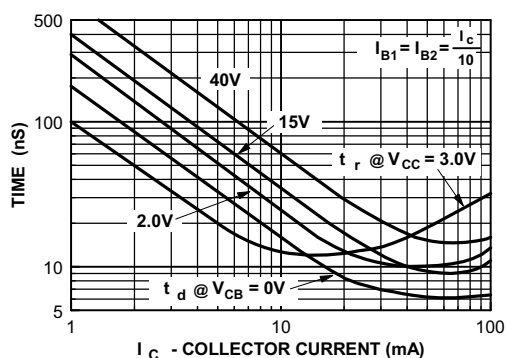


Figure 17. Turn-On Time vs. Collector Current

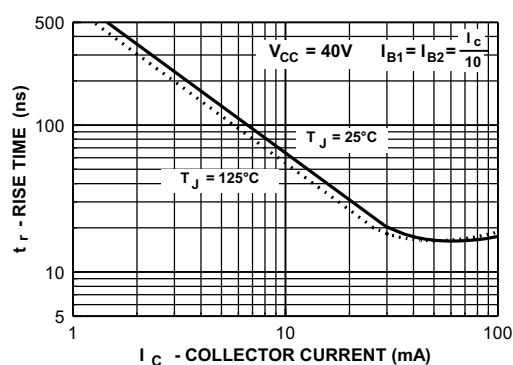


Figure 18. Rise Time vs. Collector Current

# Typical Performance Characteristics (Continued)

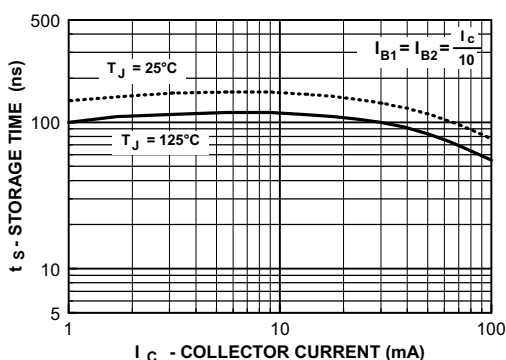


Figure 19. Storage Time vs. Collector Current

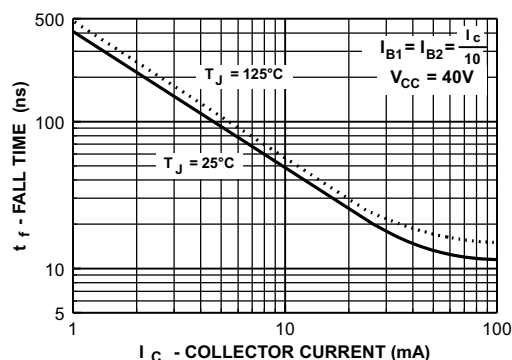


Figure 20. Fall Time vs. Collector Current

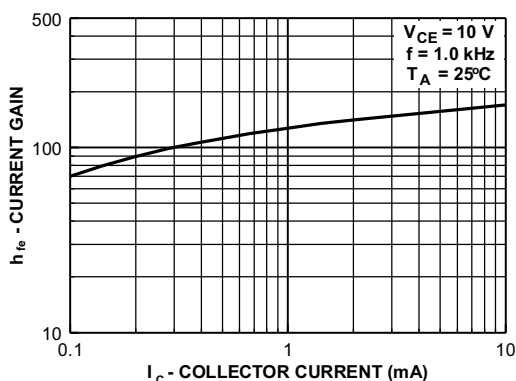


Figure 21. Current Gain

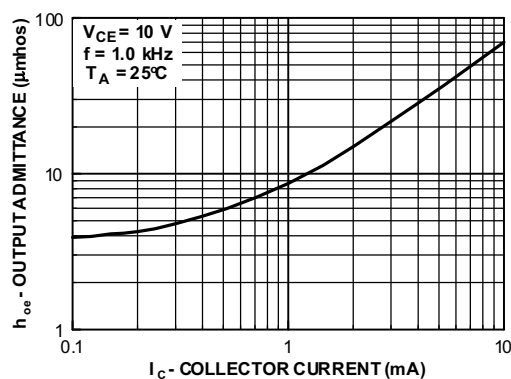


Figure 22. Output Admittance

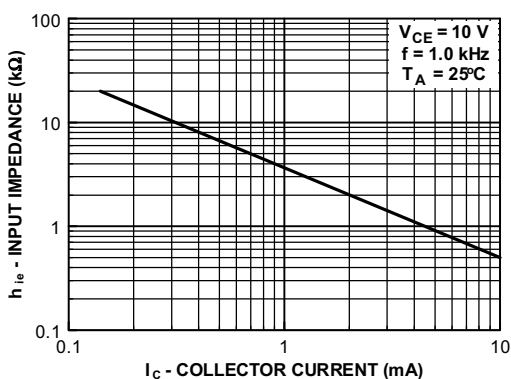


Figure 23. Input Impedance

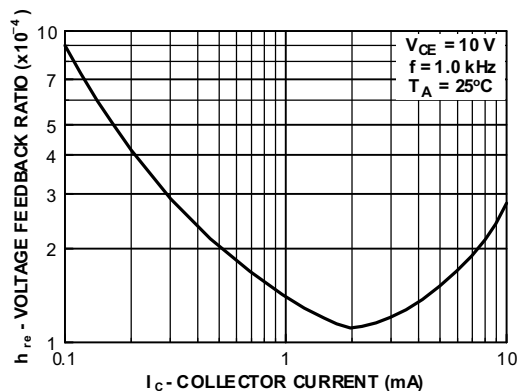
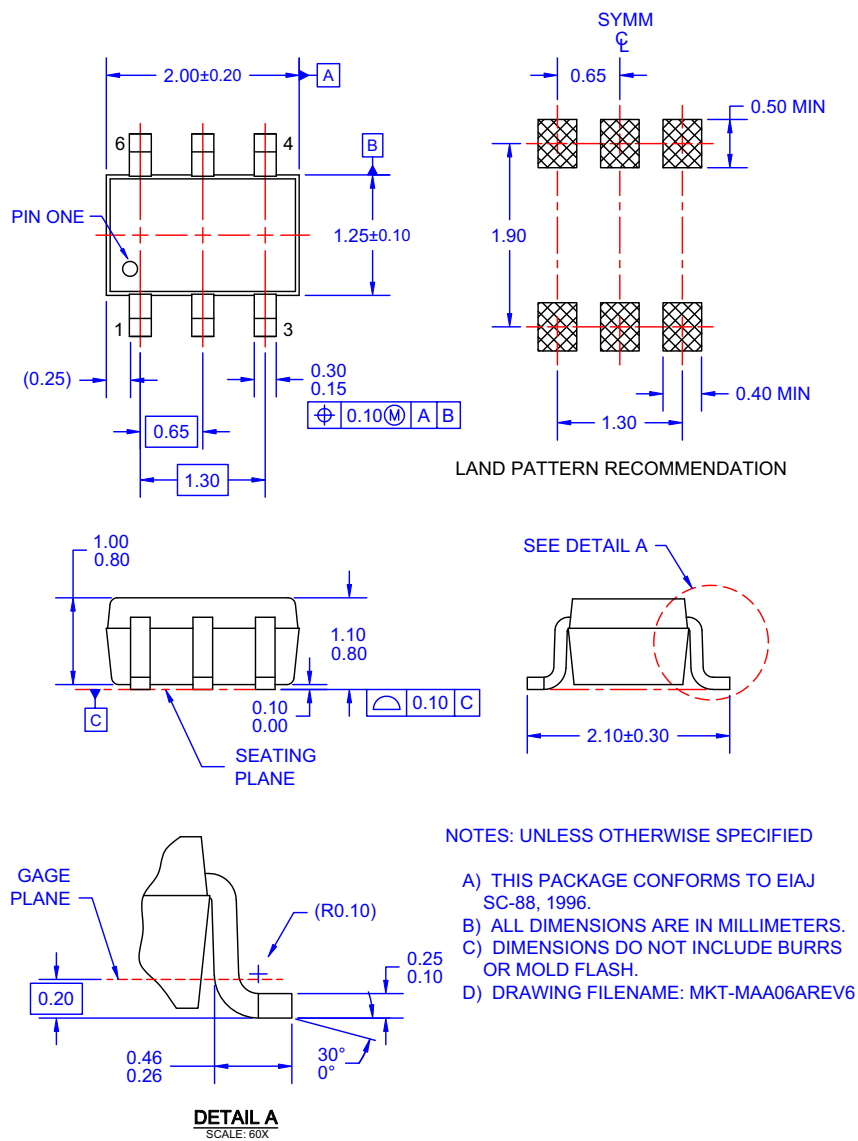


Figure 24. Voltage Feedback Ratio

## Physical Dimensions

## SC70 6L

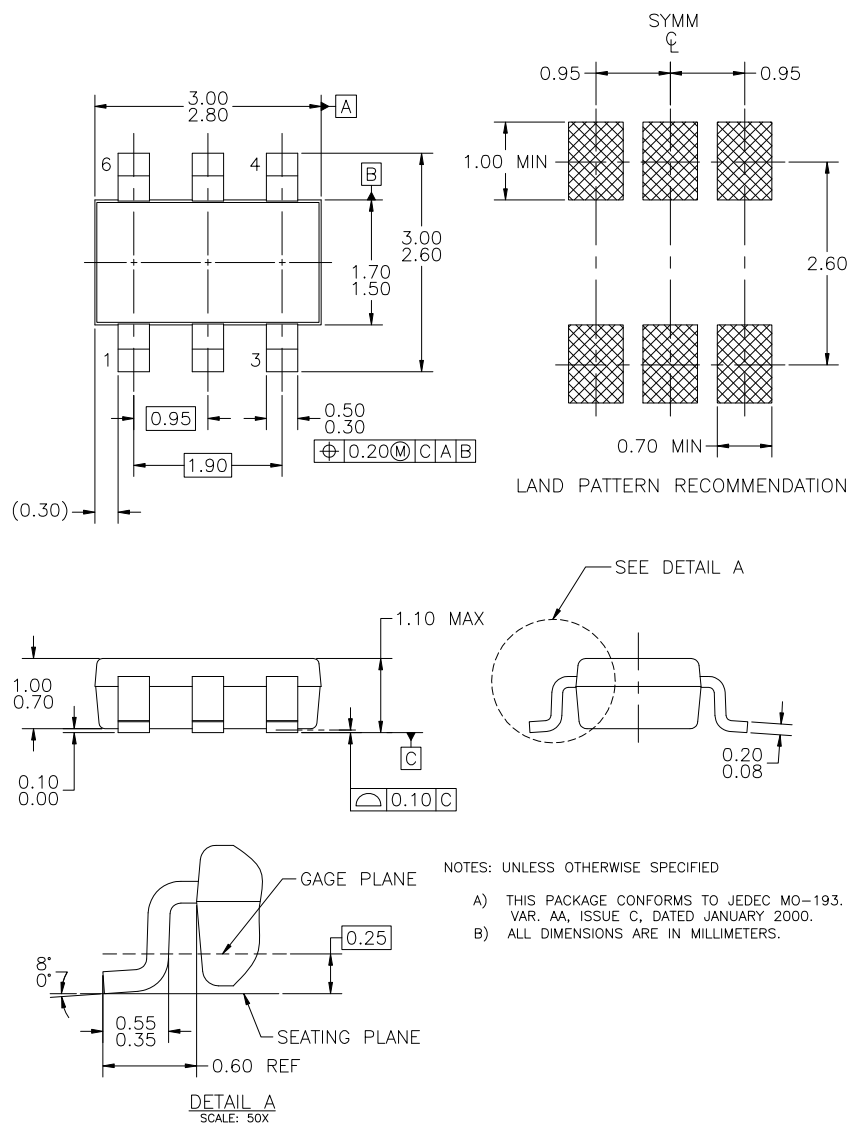


**Figure 25. 6-LEAD, SC70, EIAJ SC-88, 1.25 MM WIDE (ACTIVE)**



# Physical Dimensions (Continued)

## SSOT 6L



MA06AREVD

Figure 26. 6-LEAD, SUPERSOT-6, JEDEC MO-193, 1.6 MM WIDE (ACTIVE)

# Physical Dimensions (Continued)

## SO 16L NB

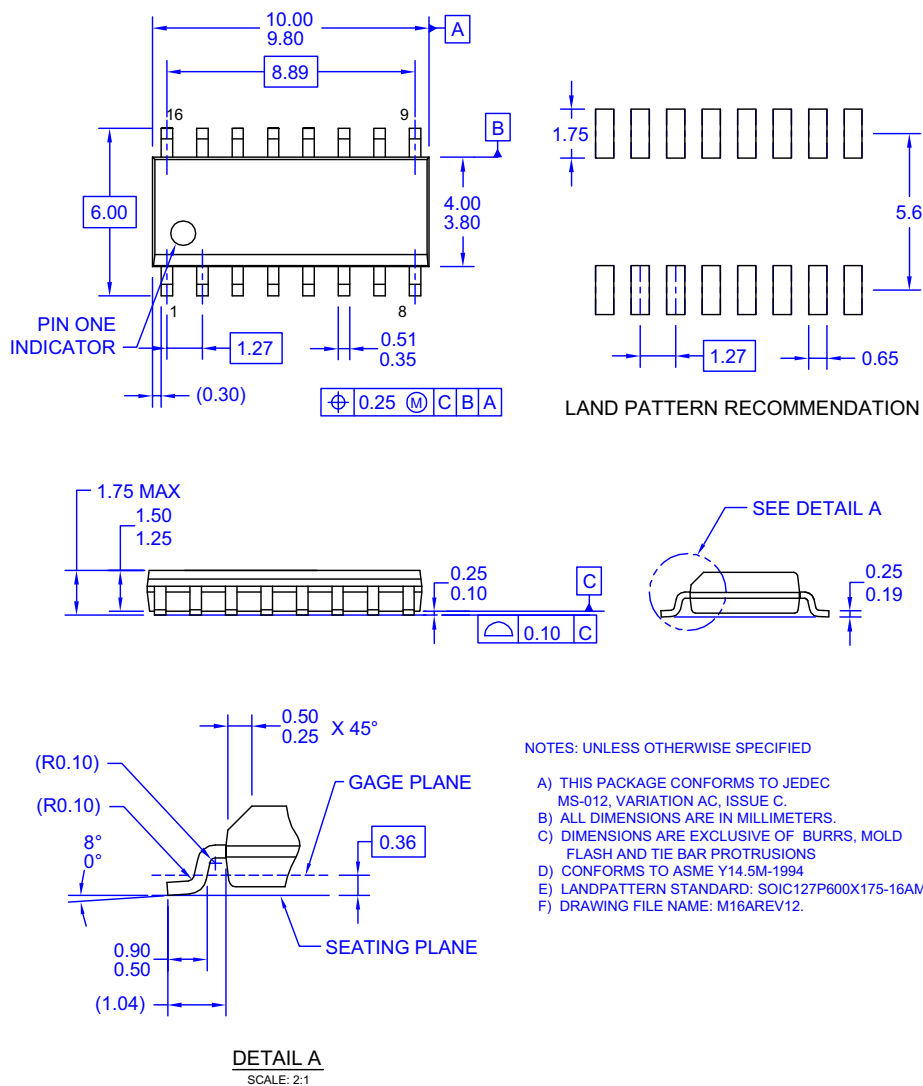


Figure 27. 16-LEAD, SOIC, JEDEC MS-012, 0.150 inch, NARROW BODY (ACTIVE)

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